Lecture 12
Time Series Analysis I

STAT 8020 Statistical Methods II

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Agenda

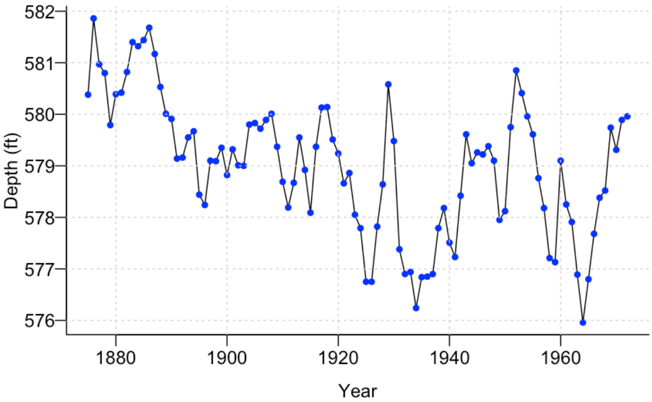
1. Background
2. Time Series Models
3. A Case Study

Level of Lake Huron 1875–1972

Annual measurements of the level of Lake Huron in feet.

[Source: Brockwell & Davis, 1991]

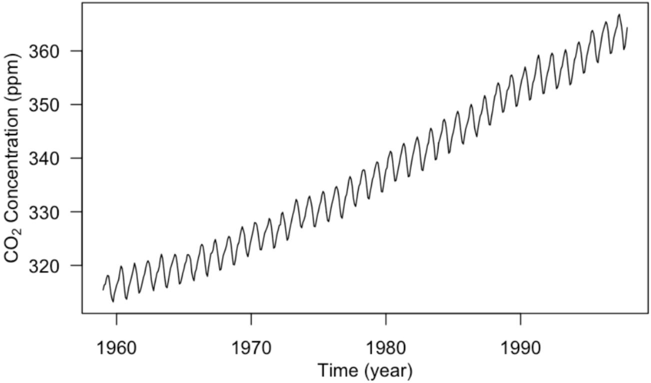
```{r}
par(mar = c(3.2, 3.2, 0.5, 0.5), mgp = c(2, 0.5, 0), bty = "L")
data(LakeHuron)
plot(LakeHuron, ylab = "Depth (ft)", xlab = "Year", las = 1)
points(LakeHuron, cex = 0.8, col = "blue", pch = 16)
grid()
```



Return

Mauna Loa Atmospheric CO_2 Concentration

Monthly atmospheric CO_2 at Mauna Loa Observatory [Source: Keeling & Whorf (SIO)]

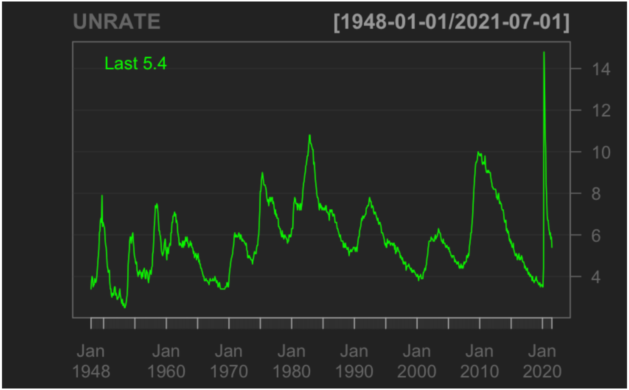


Observation

The series exhibits both a long-term increasing trend and strong yearly seasonality

US Unemployment Rate 1948 Jan. – 2021 July

[Source: St. Louis Federal Reserve Bank's FRED system]

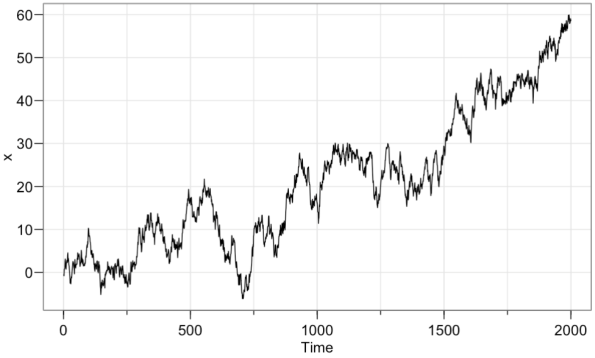


Observation

The series shows recurring economic cycles, along with occasional extreme spikes such as during COVID-19

A Simulated Time Series

```{r}
set.seed(123)
w <- rnorm(2000); x <- cumsum(w); tsplot(x, las = 1)
```



Observation

The fluctuations appear to become larger over time

Time Series Data

- A time series is a sequence of observations $\{Y_t,\ t \in T\}$ recorded over time.
  - $T = \{0, 1, 2, \ldots, T\} \subset \mathbb{Z} \Rightarrow$ discrete-time time series
  - $T = [0, T] \subset \mathbb{R} \Rightarrow$ continuous-time time series
- Discrete-time series may be naturally discrete or obtained from an underlying continuous process through:
  - **Sampling** (e.g., instantaneous wind speed)
  - **Aggregation** (e.g., daily precipitation totals)
  - **Extrema** (e.g., daily maximum temperature)
- In this course, we focus on discrete-time real-valued time series ($Y_t \in \mathbb{R}$).

Unlike independent data, nearby observations in a time series are often correlated

Exploratory Time Series Analysis

- Start with a time series plot, i.e., to plot y_t versus t
  Lake Huron Time Series
- Look at the following:
  - Are there abrupt changes?
  - Are there "outliers"?
  - Is there a need to transform the data?
- Examine the trend, seasonal components, and the "noise" term

Common Features of Time Series Data

Component	Meaning	Example
Trend (μ_t)	Long-term smooth movement in the series	Global warming
Seasonality (s_t)	Repeating pattern or cycle over time	Monthly electricity demand
Noise (η_t)	Random fluctuations not explained by trend or seasonality	Day-to-day variability

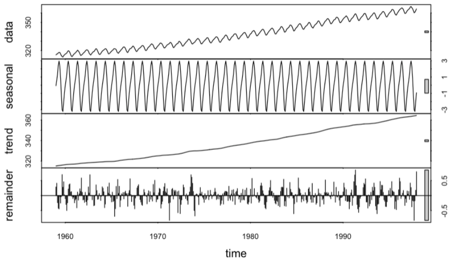
Takeaway

Many time series can be viewed as a combination of trend, seasonality, and random noise

Decomposing Time Series into Trend, Seasonality, and Noise

- Additive model:

$$y_t = \mu_t + s_t + \eta_t, \quad t = 1, \cdots, T$$



- Multiplicative model:

$$y_t = \mu_t s_t \eta_t, \quad t = 1, \ldots, T$$

If $y_t > 0$, taking logs gives an additive form:

$$\log y_t = \log \mu_t + \log s_t + \log \eta_t$$

Many time series models focus on modeling the remaining noise process after removing trend and seasonality

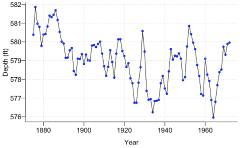
Main Goals of Time Series Analysis

Modeling

Find a statistical model that explains the observed data over time

Example: Model Lake Huron depths exhibiting

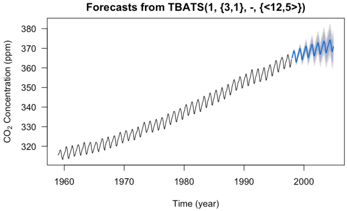
- temporal correlation
- long-term decline



Supports further inference such as: Is there evidence of a decreasing trend?

Forecasting

Use observed data to predict future values.



Common applications:

- weather
- finance
- sales
- energy demand

Additional Objectives

- **Adjustment**: an example would be seasonal adjustment, where the seasonal component is estimated and then removed in order to better understand the underlying trend
- **Simulation**: use a time series model (which adequately describes a physical process) as a surrogate to *simulate repeatedly in order to approximate how the physical process behaves*
- **Control**: adjust various input (control) parameters so that the time series fits closer to a given standard (many examples from statistical quality control)

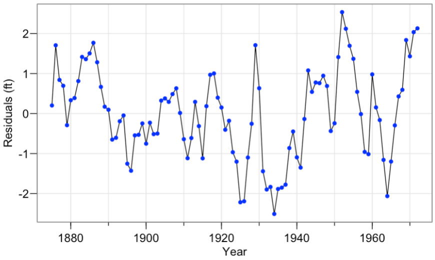
Time Series Models

Lake Huron Time Series

- Time series analysis is the area of statistics which deals with the analysis of dependency between observations over time (typically $\{\eta_t\}$)

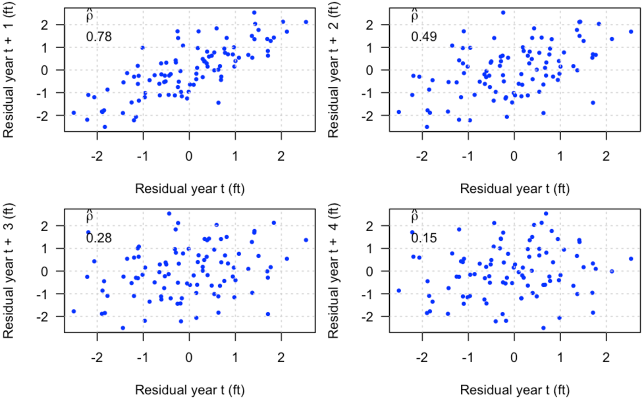
- Some key features of the Lake Huron time series:
  ▸ Lake Huron Time Series
  - decreasing trend
  - some "random" fluctuations around the decreasing trend

- We extract the "noise" component by assuming a linear trend



Exploring the Temporal Dependence Structure of $\{\eta_t\}$

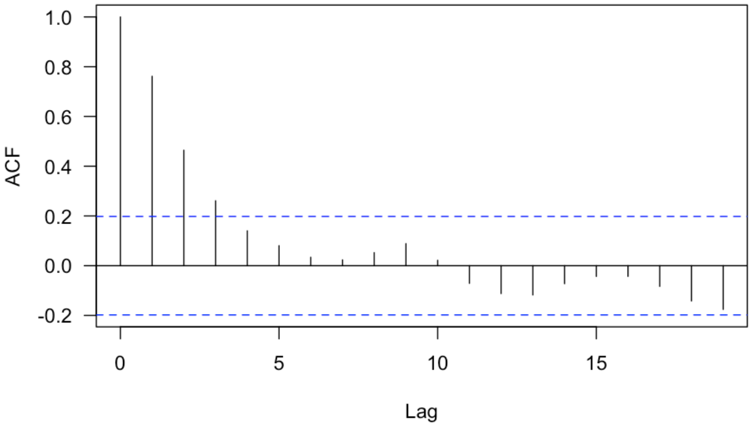
$\{\eta_t\}$ exhibit some temporal dependence structure, that is, the nearby (in time) values tend to be more alike than those far part values. To see this, let's make a few time lag plots



Positive slope in lag plots suggests positive temporal correlation

Further Exploration of the Temporal Dependence Structure

Let's plot the correlation as a function of the time lag



Positive autocorrelation persists across lags, but weakens as the lag increases. We will use this information to suggest an appropriate model

Time Series Models

- A time series model is a probabilistic model for $\{Y_t : t \in T\}$ that describes ways that the series data $\{y_t\}$ could have been generated
- Will try to keep our models for $\{Y_t\}$ simple by assuming stationarity $\Rightarrow$ characteristic of the distribution of $\{Y_t\}$ does not depend on the time points, only on the "time lag"

  We will focus on stationarity in means and autocovariances
- While most time series are not stationary, one either remove or model the non-stationary parts (e.g., de-trend or de-seasonalization) so that we are only left with a stationary component $\{\eta_t\}$.

Time Series Models

A time series model is a probabilistic model for $\{Y_t : t \in T\}$, describing how the observed series $\{y_t\}$ could have been generated

- To keep models mathematically tractable, we often assume stationarity $\Rightarrow$ distributional properties depend on lag, not on time itself

  In this course, we mainly focus on the weaker notion of stationarity based on:

  - constant means
  - autocovariances depending only on lag

- Most real-world time series are not stationary

  Common practice:
  remove or model trends/seasonality so that the remaining component $\{\eta_t\}$ is approximately stationary

Mean and Autocovariance Functions

- The mean function of $\{\eta_t\}$ is

$$\mu_t = \mathrm{E}[\eta_t], \quad t \in T$$

- The autocovariance function of $\{\eta_t\}$ is

$$\gamma(t,t') = \mathrm{Cov}(\eta_t, \eta_{t'}) = \mathrm{E}\big[(\eta_t - \mu_t)(\eta_{t'} - \mu_{t'})\big], \quad t, t' \in T$$

- When $t = t'$,

$$\gamma(t,t) = \mathrm{Var}(\eta_t) = \sigma_t^2,$$

which is the variance function of $\{\eta_t\}$.

Autocorrelation Function

The autocorrelation function (ACF) of $\{\eta_t\}$ is

$$\rho(t,t') = \mathrm{Corr}(\eta_t, \eta_{t'}) = \frac{\gamma(t,t')}{\sqrt{\gamma(t,t)\gamma(t',t')}}$$

It measures the strength of linear association between η_t and $\eta_{t'}$

Properties:

1. $-1 \le \rho(t,t') \le 1, \quad t,t' \in T$
2. $\rho(t,t') = \rho(t',t), \quad \forall t,t' \in T; \rho(t,t) = 1, \quad \forall t \in T$
3. $\rho(t,t')$ is a non-negative definite function

Partial autocorrelation function (PACF) is a conditional correlation, i.e., the correlation at two time points given the information at all other time points

Weakly Stationary Processes

To simplify modeling, we often assume weak stationarity, meaning that the distributional properties depend only on the time lag

Weak stationarity requires:

- Constant mean:

$$E[\eta_t] = \mu, \quad \forall t \in T$$

- Autocovariance depends only on lag:

$$\mathrm{Cov}(\eta_t, \eta_{t'}) = \gamma(t' - t) = \gamma(h)$$

  Equivalently,

$$\mathrm{Cov}(\eta_t, \eta_{t'}) = \mathrm{Cov}(\eta_{t+s}, \eta_{t'+s})$$

Under stationarity, the ACF depends only on lag:

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}$$

Autoregressive Moving Average (ARMA) Models

Let $\{Z_t\}$ be independent and identical random variables that follow $\mathrm{N}(0, \sigma^2)$

- Moving Average Processes (MA(q)):

$$\eta_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} \cdots + \theta_q Z_{t-q}$$

$\Rightarrow$ Current values depend on previous random shocks

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- Autoregressive Processes (AR(p)):

$$\eta_t = \phi_1 \eta_{t-1} + \phi_2 \eta_{t-2} + \cdots + \phi_p \eta_{t-p} + Z_t$$

$\Rightarrow$ Current values depend on previous observations

Autoregressive Moving Average (ARMA) Models

Let $\{Z_t\}$ be independent and identical random variables that follow $\mathrm{N}(0, \sigma^2)$

- Moving Average Processes (MA(q)):

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$\Rightarrow$ Current values depend on previous observations

- Autoregressive Moving Average Processes ARMA(p,q):

$$\eta_t = \phi_1 \eta_{t-1} + \phi_2 \eta_{t-2} + \cdots + \phi_p \eta_{t-p} + Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \cdots + \theta_q Z_{t-q}$$

$\Rightarrow$ Combines dependence on past observations and past shocks

AR vs. MA vs. ARMA Models

Model	Depends On
AR	Past observations
MA	Past errors/shocks
ARMA	Both

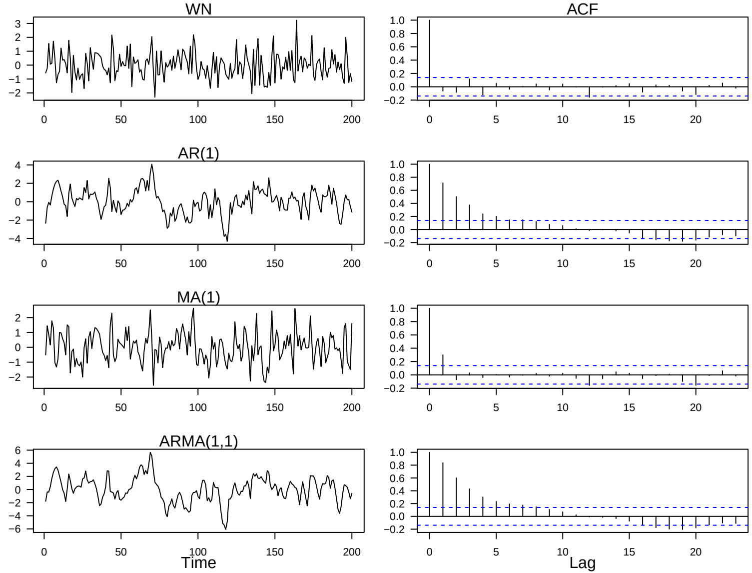
Examples:

AR(1): $\eta_t = \phi\eta_{t-1} + Z_t$

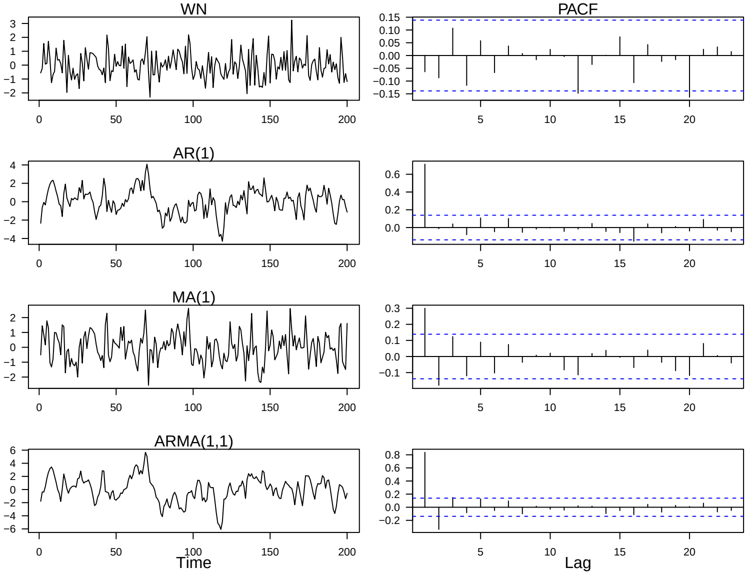
MA(1): $\eta_t = Z_t + \theta Z_{t-1}$

ARMA(1,1): $\eta_t = \phi\eta_{t-1} + Z_t + \theta Z_{t-1}$

ACF Plots



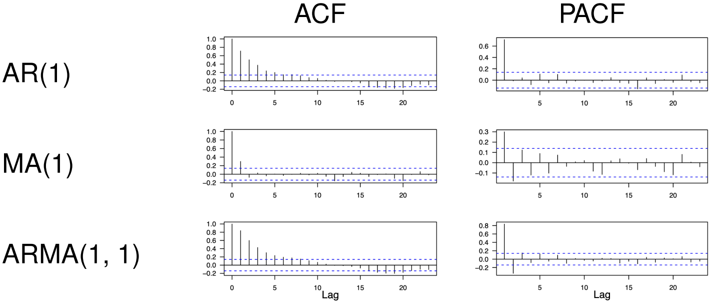
PACF Plots



Identification of ARMA Models using ACF/PACF Plots

Use the ACF and PACF together to identify candidate models. The following table gives some rough guidelines.

	ACF	PACF
AR(p)	Tails off	Cuts off after lag p
MA(q)	Cuts off after lag q	Tails off
ARMA(p, q)	Tails off	Tails off



Remark: ACF/PACF identification is often heuristic and mainly used to suggest initial candidate models; **criteria such as AIC or BIC are then used for formal model selection**

Model Diagnostics: Ljung-Box Test [Ljung and Box, 1978]

We wish to test:

$H_0 : \{e_1, e_2, \cdots, e_T\}$ is an i.i.d. noise sequence $\Rightarrow$ model adequate
$H_1 : H_0$ is false $\Rightarrow$ model not good,

where $\{e_t\}$ are the residuals after fitting a model to $\{\eta_t\}$

Test statistic:

$$Q_{LB} = T(T-2)\sum_{h=1}^{\text{lag}} \frac{\hat{\rho}^2_{\hat{\boldsymbol{e}}}(h)}{T-h} \overset{H_0}{\approx} \chi^2_k,$$

where T is the sample size, $\hat{\rho}_{\hat{\boldsymbol{e}}}(h)$ is the sample ACF at lag h, applied to the residuals of a fitted ARIMA model. The degrees of freedom $k = \text{Lag} - p - q$.

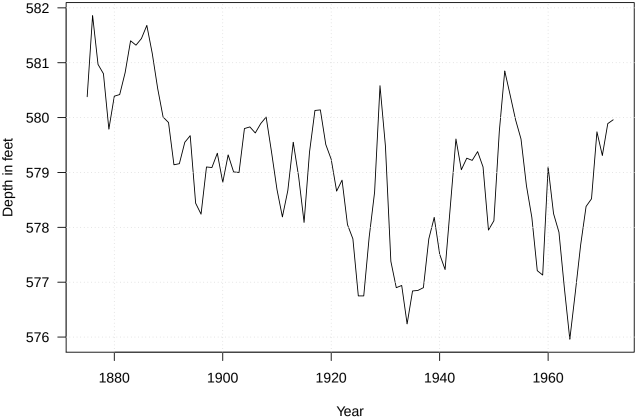
Ljung-Box test can be carried out in `R` using the function `Box.test`

Lake Huron Case Study

Source: `https://www.worldatlas.com/articles/what-states-border-lake-huron.html`

- Detrending
- Model fitting and selection
- Forecasting

Annual Measurements of the Level of Lake Huron

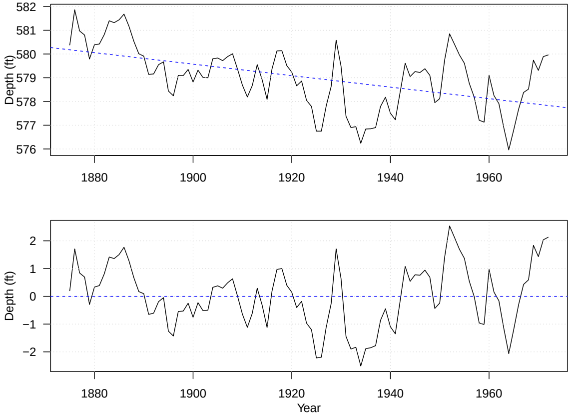


There seems to be a decreasing trend ⇒ need to estimate the trend to get the detrended series

Plots of the Trend and Residuals

$$y_t = \overbrace{\mu_t}^{\text{trend}} + \underbrace{\eta_t}_{\text{residual}},$$

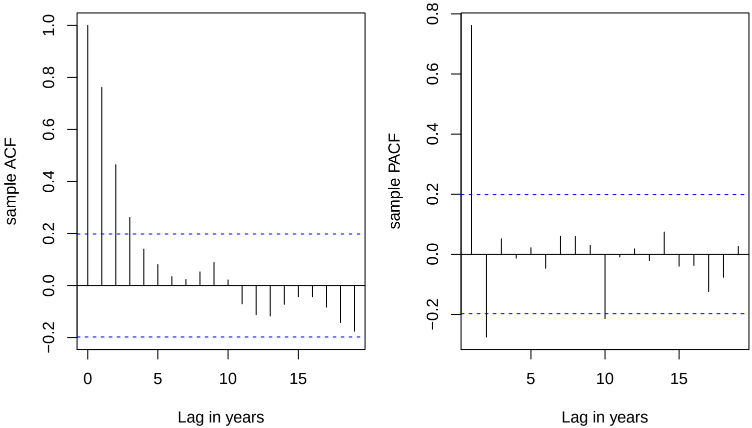
where we **assume** $\mu_t = \alpha + \beta t$, i.e., a linear trend in time



Return

ACF and PACF Plots

- Tapering pattern in ACF ⇒ need to include AR terms
- Significant PACF values at the first 2 lags ⇒ an AR(2) candidate model



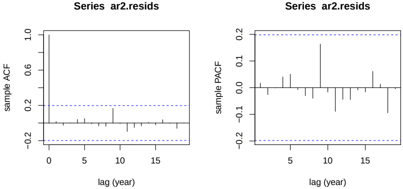
Fitting an AR(2) to the Detrended Time Series

```
> (ar2.model <- arima(deTrend, order = c(2, 0, 0), method = "ML"))

Call:
arima(x = deTrend, order = c(2, 0, 0), method = "ML")

Coefficients:
         ar1      ar2  intercept
      1.0047  -0.2919     0.0197
s.e.  0.0977   0.1004     0.2350

sigma^2 estimated as 0.4571:  log likelihood = -101.25,  aic = 210.5
```

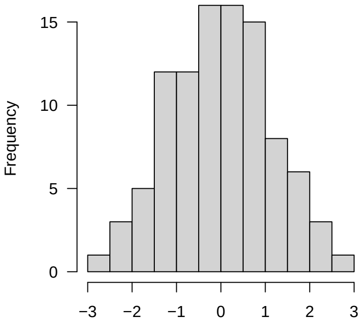


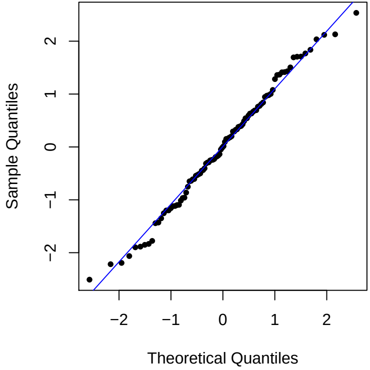

```
> Box.test(ar2.resids, lag = 5, fitdf = 2, type = "Ljung-Box")

        Box-Ljung test

data:  ar2.resids
X-squared = 0.55962, df = 3, p-value = 0.9056
```

Assessing Normality Assumption for η_t





- Histogram: To compare the shape of the distribution of residuals with the bell-shaped normal density curve
- Q-Q plot: To compare the quantiles of the residual distribution to the quantiles of a normal distribution

Model Selection via AIC

We can conduct model selection by using, for example, AIC

```
> auto.arima(deTrend, trace = T)

 ARIMA(2,0,2) with non-zero mean : 215.0455
 ARIMA(0,0,0) with non-zero mean : 304.222
 ARIMA(1,0,0) with non-zero mean : 216.8388
 ARIMA(0,0,1) with non-zero mean : 235.4585
 ARIMA(0,0,0) with zero mean     : 302.1373
 ARIMA(1,0,2) with non-zero mean : 212.7747
 ARIMA(0,0,2) with non-zero mean : 218.2478
 ARIMA(1,0,1) with non-zero mean : 210.9477
 ARIMA(2,0,1) with non-zero mean : 212.8306
 ARIMA(2,0,0) with non-zero mean : 210.9333
 ARIMA(3,0,0) with non-zero mean : 212.7787
 ARIMA(3,0,1) with non-zero mean : Inf
 ARIMA(2,0,0) with zero mean     : 208.7655
 ARIMA(1,0,0) with zero mean     : 214.7735
 ARIMA(3,0,0) with zero mean     : 210.569
 ARIMA(2,0,1) with zero mean     : 210.6186
 ARIMA(1,0,1) with zero mean     : 208.7891
 ARIMA(3,0,1) with zero mean     : Inf

 Best model: ARIMA(2,0,0) with zero mean
```

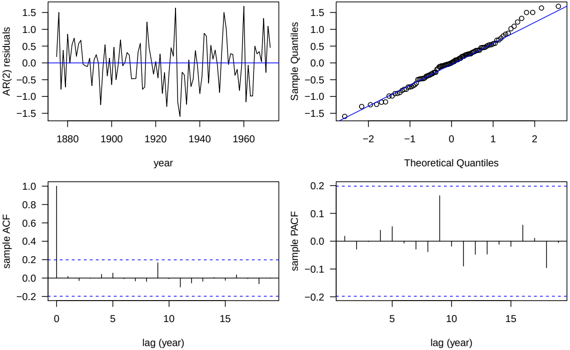
Smaller AIC values indicate better tradeoff between model fit and model complexity

Fitting Linear Trend and ARMA in One Step

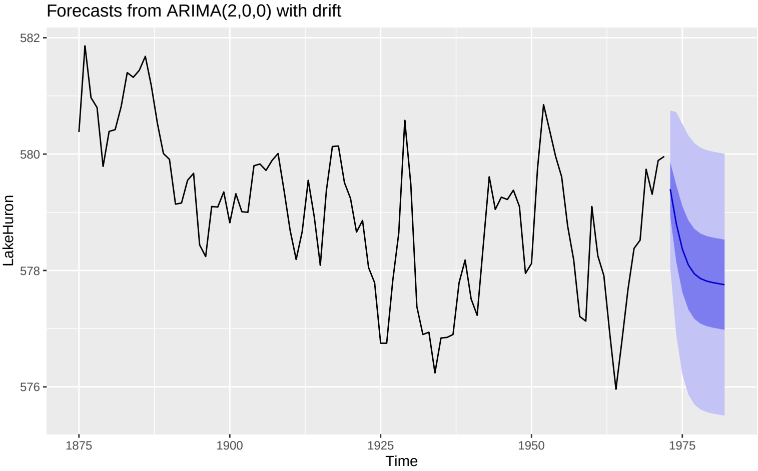
```
> (fit <- Arima(LakeHuron, order = c(2, 0, 0), include.drift = T))
Series: LakeHuron
ARIMA(2,0,0) with drift

Coefficients:
         ar1      ar2  intercept    drift
      1.0048  -0.2913   580.0915  -0.0216
s.e.  0.0976   0.1004     0.4636   0.0081

sigma^2 = 0.476:  log likelihood = -101.2
AIC=212.4   AICc=213.05   BIC=225.32
```



10-Year-Ahead Forecasts



Prediction intervals widen over time because uncertainty accumulates further into the future

Summary

These slides cover:

- Basic concepts of time series analysis
- A widely used class of models: ARMA
- ARMA model identification, estimation/prediction, inference

`R` functions to know:

- `acf` and `pacf` for identifying candidate models
- `arima` and `Arima` (under the package `forecast`) for model fitting
- `auto.arima` for model selection
- `Box.test` for testing model adequacy
- `forecast` (under the package `forecast`) for generating forecasts and prediction intervals